User Guide for STRATIGIRAFFIC

Quick Start

(You can follow the steps below while referring to the figures at the bottom of the page)

1) Install: run SGF_Installer_web.exe (see the left figure below for the installer interface). Follow the on-screen instructions to install SGF. Don't forget to create a desktop shortcut for quick and convenient access.

If Windows blocks the installer, choose More info → Run anyway (or the equivalent option) to proceed.

2) Launch: Open the created shortcut for STRATIGIRAFFIC.exe on the desktop (see the right figure below for the main interface).

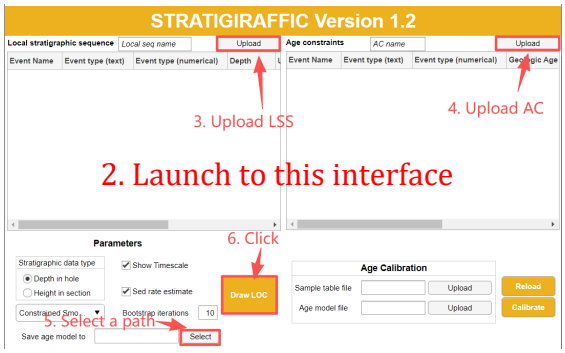
3) Upload Local Stratigraphic Sequence table (LSS): click 'Upload' button and select your Local Stratigraphic Sequence file (Use 'LSS-demo.csv' provided in the demo folder 'SGF 20260109 with demo' or you own table in the same format as 'LSS-demo.csv').

4) Upload Age Constraints table (AC): click 'Upload' button and select your Age Constraints table (Use 'AC-demo.xlsx' provided in the demo folder 'SGF 20260109 with demo' or you own table in the same format as 'AC-demo.xlsx').

5) Select a path that you 'Save age model to'.

6) Click 'Draw LOC' button: the age–depth plot will be generated and the calibration table will be exported/updated automatically.

If this is your first run, a parallel acceleration service may start automatically; it speeds up subsequent runs but may ***take a few minutes****.*



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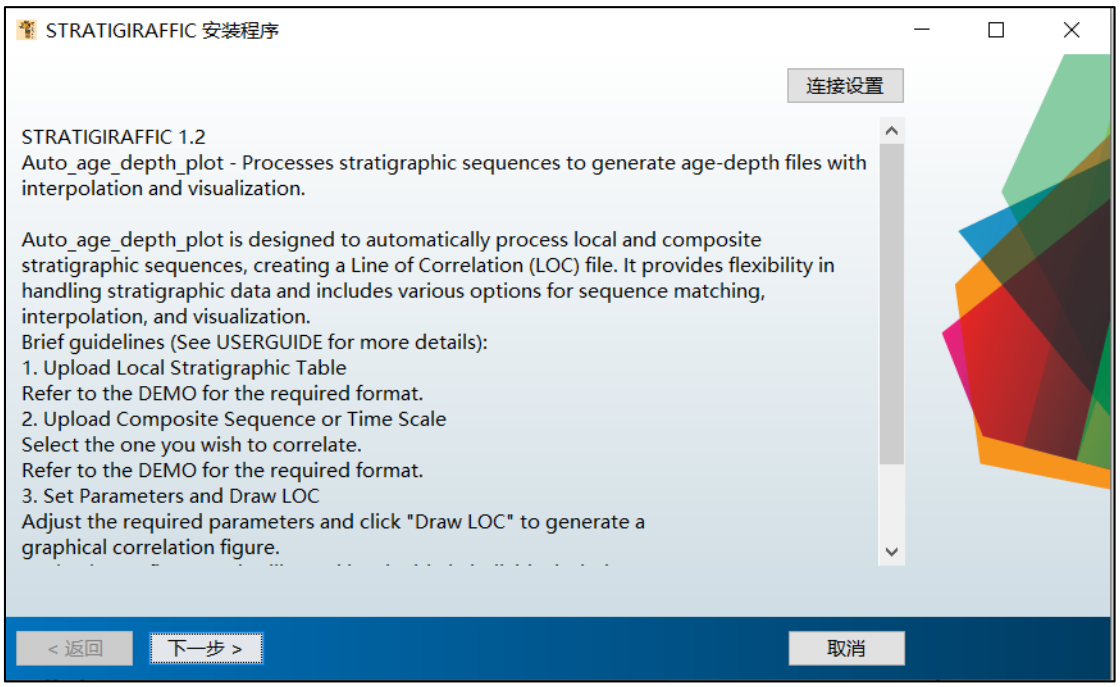
Function 1: Construct an Age Model with SGF

Install the application

Run SGF_Installer_web.exe from this folder.

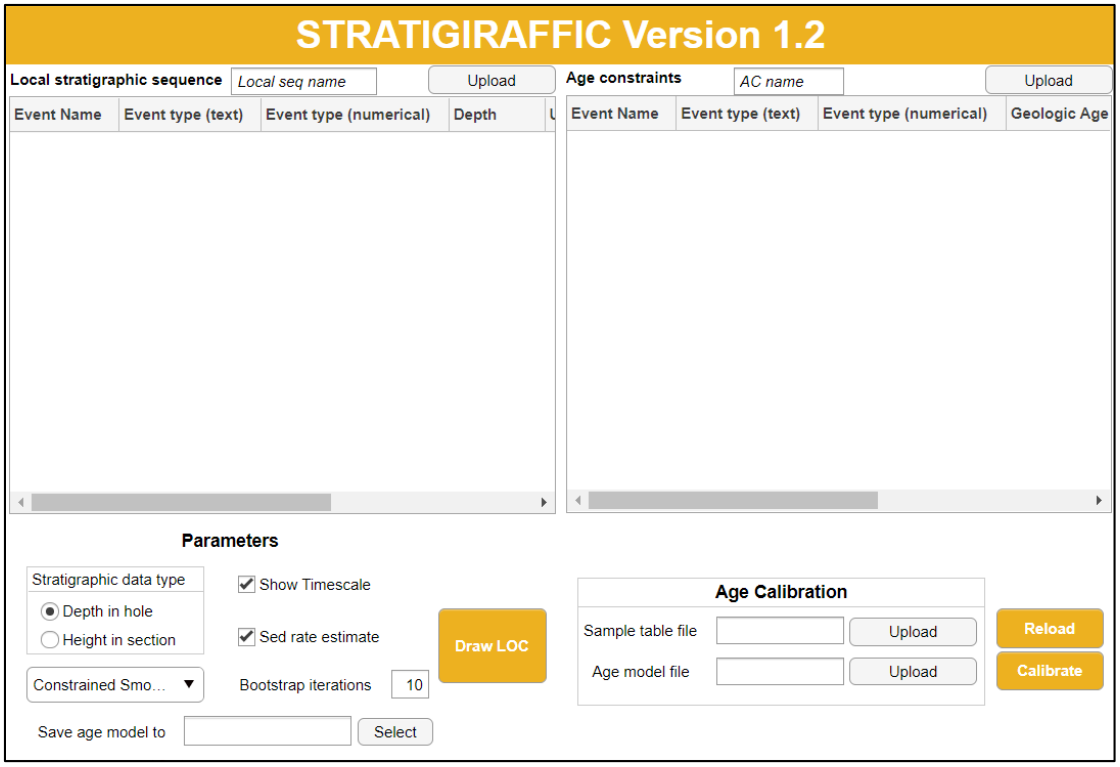
If Windows displays a security prompt (e.g., “Unknown publisher”), choose Run anyway / More info → Run anyway, and avoid interrupting the installation.

Make sure all required dependencies are installed when prompted.



Launch the app

Open STRATIGIRAFFIC.exe.



Upload a Local Stratigraphic Sequence (LSS)

Upload your local stratigraphic sequence using the same format as the example file LSS-demo.csv.

The LSS table contains seven columns describing stratigraphic horizons (depth/height attributes):

• Event Name: Name of the datum (e.g., T. *Globigerinatheka index*). Use the same name in the Age Constraints table (AC) for the same horizon.

• Event type (text): A descriptive label (e.g., PF = Planktonic Foraminifera). This is for display/legend purposes and can be customized.

• Event type (numerical): Numeric code describing the horizon type. Use the same code in the AC table for the same horizon. See table below for codes.

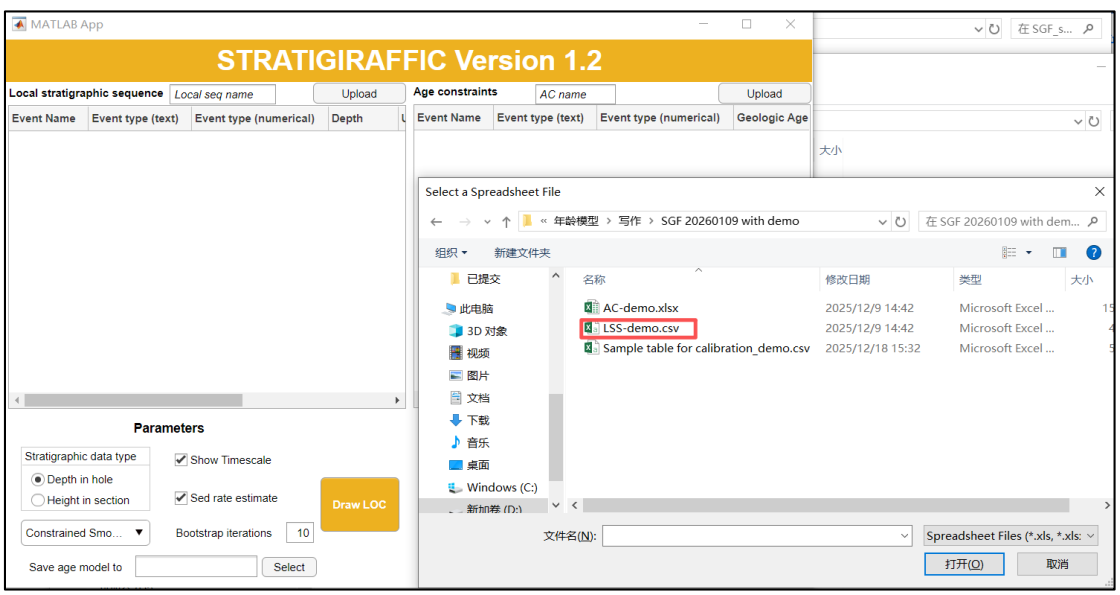
Code	Event type	Details
-3	Event bed	A discrete sedimentary layer, such as an ash bed or turbidite, that represents a short-lived depositional event
-2	Stratigraphic disturbance	An interval of stratigraphy disturbed in the later geologic process
-1	Hiatus	A break or gap in the geological record, representing an interval of missing time that is not recorded in the sedimentary succession
1	Base	Lowest observation of a fossil in the stratigraphic sequence
2	Top	Highest observation of a fossil in the stratigraphic sequence
3	Pmag	Paleomagneto reversal
4	Depth maximum event	Lowest depth estimation of an event
5	Depth minimum event	Highest depth estimation of an event
6	Top or Base of stratigraphic sequence	A known top or bottom of a given stratigraphic sequence
7	Radio-isotopic dating	Numerical age determination of stratigraphic layers using radiometric methods such as U–Pb or K–Ar dating.
8	Astro-cyclo	Tie-point markers based on astronomical (orbital) cycles.
10	Litho-bound	A known boundary of litho-unit shift
11	Undefined event	Used for any tie point type that is not specifically defined.
14	Age maximum event	Oldest age estimate of an event
15	Age minimum event	Youngest age estimate of an event

• Depth: Depth (or height) of the horizon in the local sequence.

• Uncertainty_neg: Downward (negative) depth uncertainty.

• Uncertainty_pos: Upward (positive) depth uncertainty.

• Score: Confidence/weight of the depth placement (typically 0–10; higher = more reliable). Use -1 to indicate infinite.



Upload an Age Constraints table (AC)

Upload the age-constraint table you want to correlate with, using the same format as the example file AC-demo.xlsx.

The AC table contains seven columns describing age attributes of stratigraphic tie points:

• Event Name: Tie-point name. Must match the corresponding horizon name in the LSS table.

• Event type (text): Descriptive label (e.g., PF).

• Event_Code: Numeric horizon-type code. Must match the corresponding code in the LSS table.

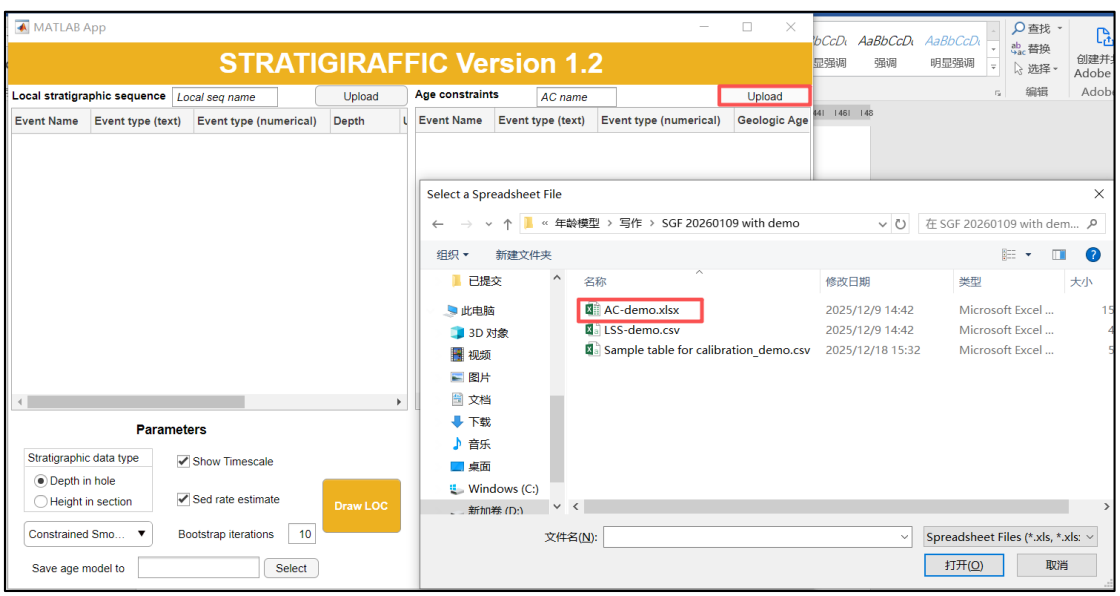
• Geologic Age: Calibrated age of the horizon/tie point.

• Uncertainty_neg: Negative (older) age uncertainty.

• Uncertainty_pos: Positive (younger) age uncertainty.

• Score: Confidence/weight of the age calibration (typically 0–10; higher = more reliable). Use -1 to indicate infinite.

Important: for the same stratigraphic horizon, ensure the Event Name and Event type (numerical) are identical between the LSS and AC tables.



Set parameters

Select the stratigraphic type of the local sequence: Core or Section (outcrop) (this controls the plot layout).

Choose the fitting method from the dropdown menu. Constrained Smoothing Spline – GCV is recommended in most cases.

Set Bootstrap Iterations to estimate uncertainty in calibrated ages (propagating uncertainties from both tables).

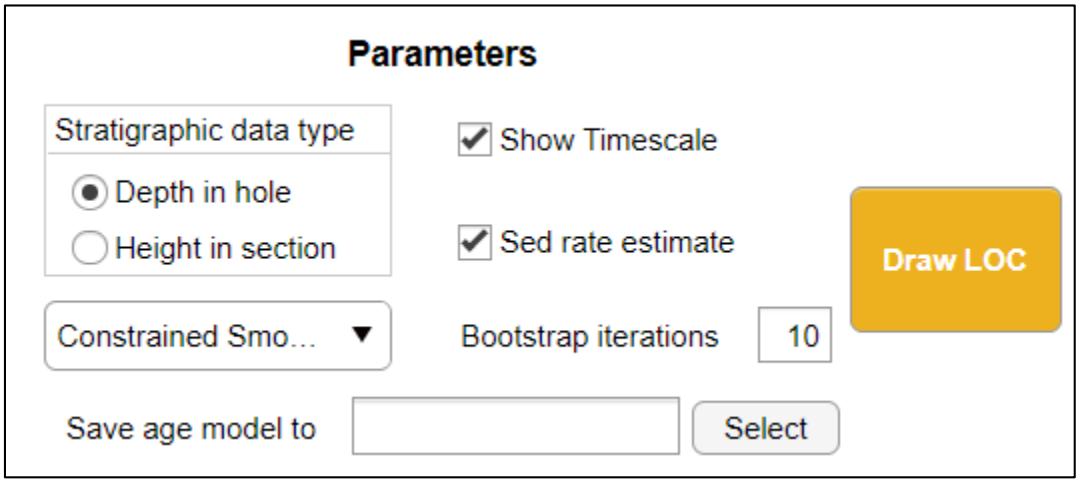
- For robust uncertainty estimates, ≥200 iterations is recommended.

- For a quick preview, start with a small number (e.g., 10) to check that inputs and settings behave as expected.

Tick Show Timescale to add timescale boxes (epoch/stage) and polarity reversals (Cenozoic paleomagnetism currently supported).

Tick Sed rate estimate to display a sedimentation-rate subplot.

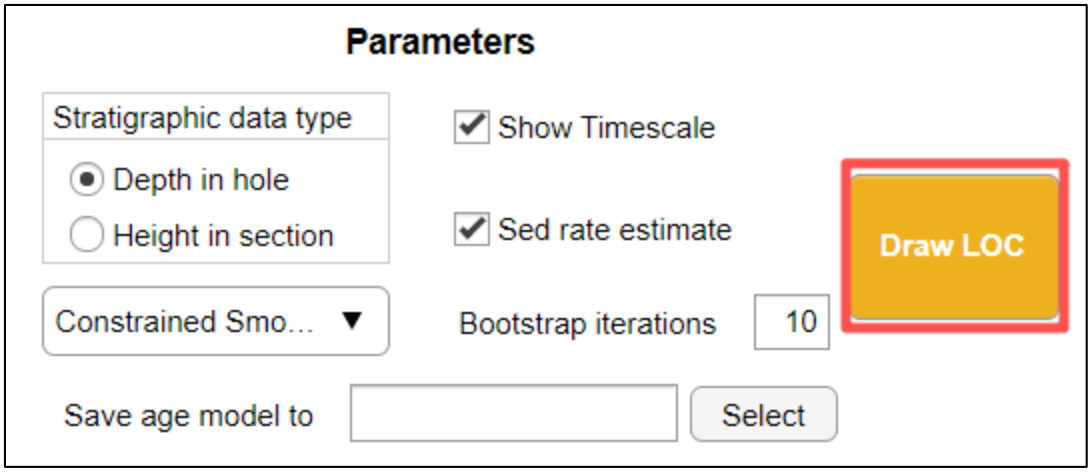
Choose an Output path to save the age-model file. This file stores inputs, fitted functions, and outputs, and can be reloaded or shared.



Draw the age–depth plot and run age calibration

Click Draw LOC to generate the age–depth plot.

On the first run, a parallel acceleration service may start automatically. This may take a few minutes the first time, but will significantly speed up subsequent calculations. You do not need to start it manually during routine use; it may shut down automatically after a long idle period.



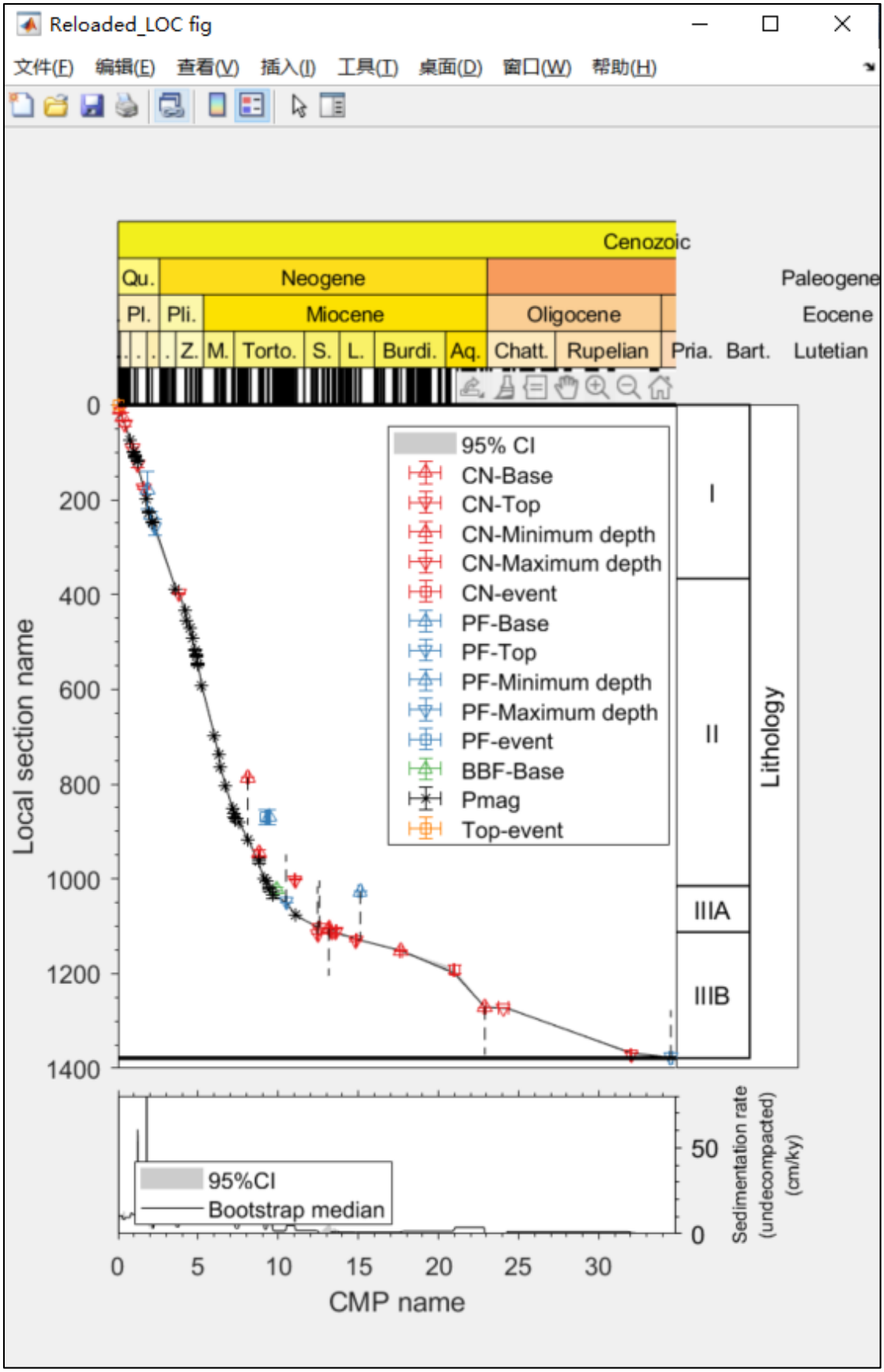
Outputs

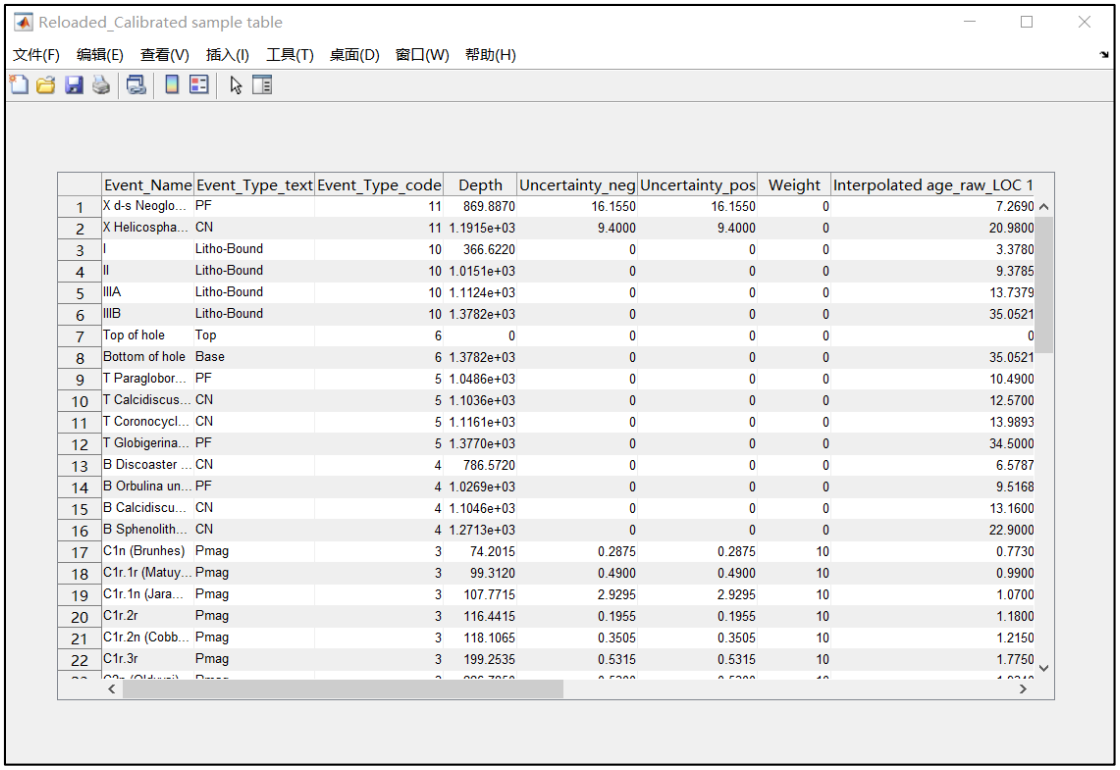
Age–depth model figure: displayed in a separate interactive figure window. You can pan/zoom, drag items, and click data points to view details.

Age calibration table: generated automatically using the fitted age–depth relationship to interpolate ages for events in the LSS.

- Calibrated ages are appended as one or more new columns (depending on how many age–depth plots are produced).
- The on LOC column indicates whether an event lies on the LOC (1 = yes, 0 = no).
- The calibration method is documented in Note_LOC.
- If Bootstrap Iterations > 0, additional columns are appended with uncertainty summaries (e.g., 95% and 99% confidence intervals and the median of the bootstrap distribution).

You can export outputs by saving/copying from the figure and table windows in your preferred formats.


Reloaded_LOC fig
文件(F) 编辑(E) 查看(V) 插入(I) 工具(T) 桌面(D) 窗口(W) 帮助(H)
Cenozoic
Qu.
Neogene
Paleogene
Pl.
Pli.
Miocene
Oligocene
Eocene
Z.
M.
Torto.
S.
L.
Burdi.
Aq.
Chatt.
Rupelian
Pria.
Bart.
Lutetian
95% CI
CN-Base
CN-Top
CN-Minimum depth
CN-Maximum depth
CN-event
PF-Base
PF-Top
PF-Minimum depth
PF-Maximum depth
PF-event
BBF-Base
Pmag
Top-event
I
II
IIIA
IIIB
Lithology
Local section name
0
200
400
600
800
1000
1200
1400
95%CI
Bootstrap median
50
0
Sedimentation rate (undecompacted) (cm/ky)
0
5
10
15
20
25
30
CMP name



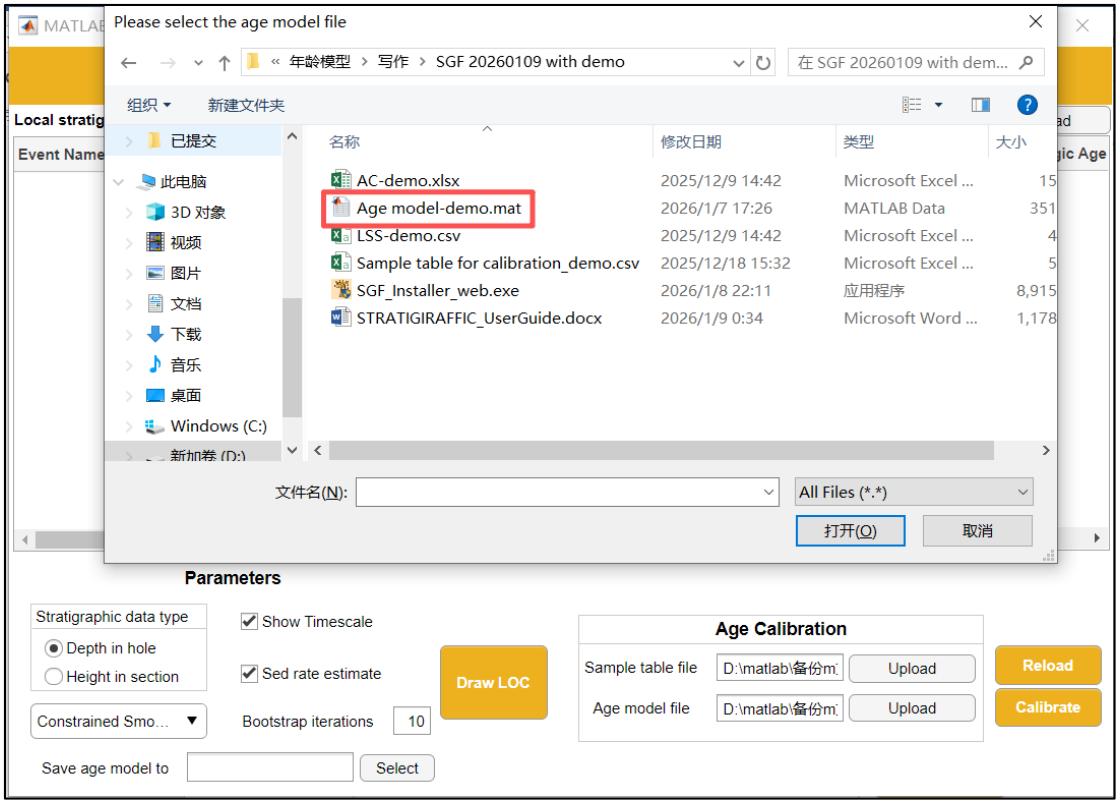
	Event_Name	Event_Type_text	Event_Type_code	Depth	Uncertainty_neg	Uncertainty_pos	Weight	Interpolated age_raw_LOC 1
1	X d-s Neoglo...	PF	11	869.8870	16.1550	16.1550	0	7.2690
2	X Helicospha...	CN	11	1.1915e+03	9.4000	9.4000	0	20.9800
3	I	Litho-Bound	10	366.6220	0	0	0	3.3780
4	II	Litho-Bound	10	1.0151e+03	0	0	0	9.3785
5	IIIA	Litho-Bound	10	1.1124e+03	0	0	0	13.7379
6	IIIB	Litho-Bound	10	1.3782e+03	0	0	0	35.0521
7	Top of hole	Top	6	0	0	0	0	0
8	Bottom of hole	Base	6	1.3782e+03	0	0	0	35.0521
9	T Paraglobor...	PF	5	1.0486e+03	0	0	0	10.4900
10	T Calcidiscus...	CN	5	1.1036e+03	0	0	0	12.5700
11	T Coronocycl...	CN	5	1.1161e+03	0	0	0	13.9893
12	T Globigerina...	PF	5	1.3770e+03	0	0	0	34.5000
13	B Discoaster ...	CN	4	786.5720	0	0	0	6.5787
14	B Orbulina un...	PF	4	1.0269e+03	0	0	0	9.5168
15	B Calcidiscu...	CN	4	1.1046e+03	0	0	0	13.1600
16	B Sphenolith...	CN	4	1.2713e+03	0	0	0	22.9000
17	C1n (Brunhes)	Pmag	3	74.2015	0.2875	0.2875	10	0.7730
18	C1r.1r (Matuy...	Pmag	3	99.3120	0.4900	0.4900	10	0.9900
19	C1r.1n (Jara...	Pmag	3	107.7715	2.9295	2.9295	10	1.0700
20	C1r.2r	Pmag	3	116.4415	0.1955	0.1955	10	1.1800
21	C1r.2n (Cobb...	Pmag	3	118.1065	0.3505	0.3505	10	1.2150
22	C1r.3r	Pmag	3	199.2535	0.5315	0.5315	10	1.7750

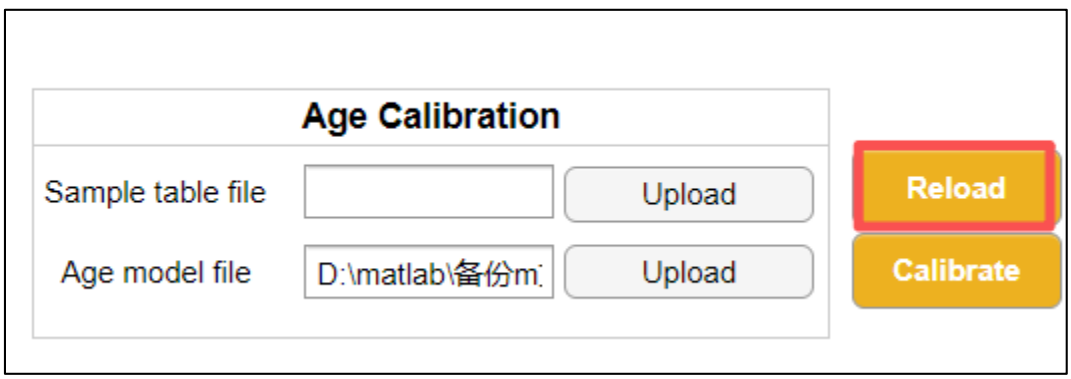
Function 2: Reload and Reuse an Existing Age Model

Reload an existing age model

Upload an age-model file generated by STRATIGIRAFFIC (or shared by another user).

The app will reload the age–depth model figure and the associated calibrated table.





Calibrate a sample table using an existing age model

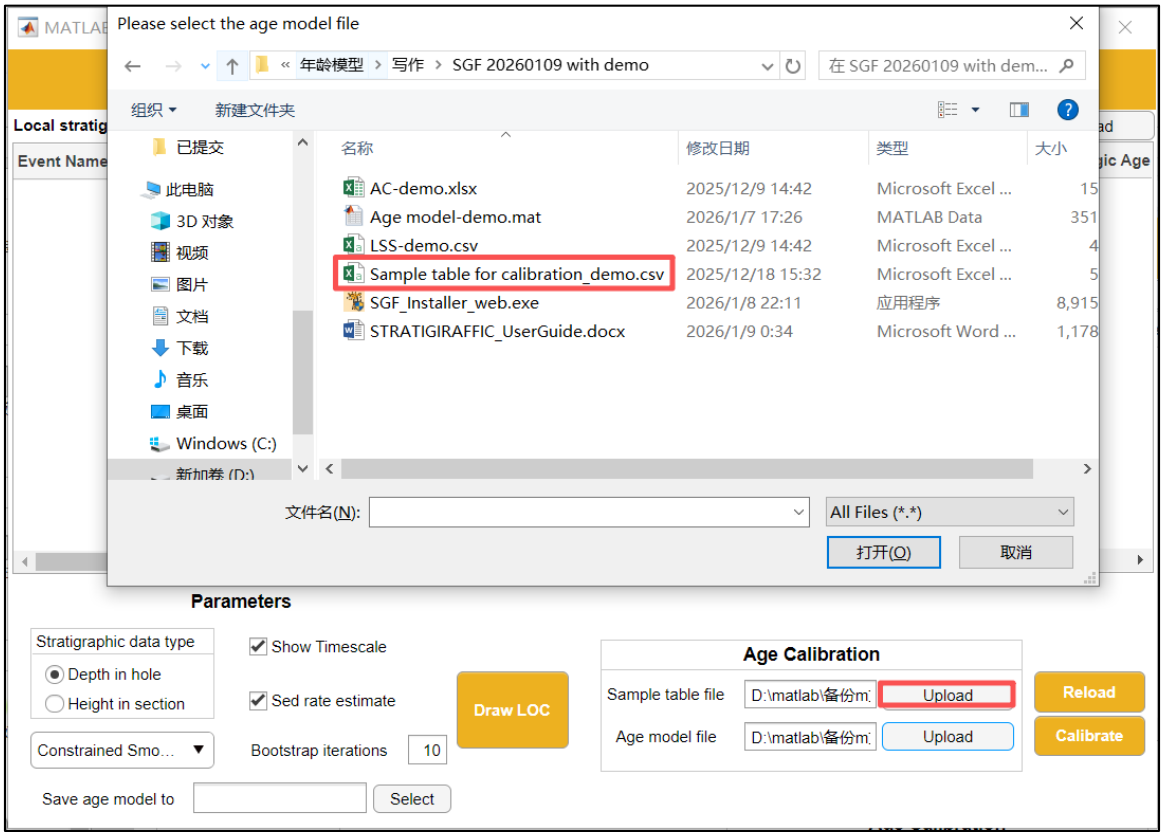
Upload demo sample table 'Sample table for calibration_demo.xlsx' or your Sample table in the same structure. The table may have any structure, provided that:

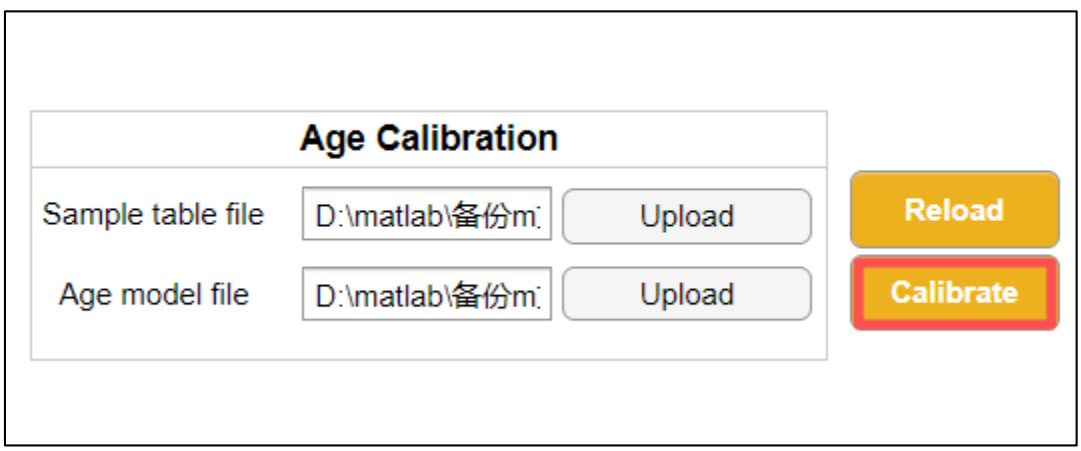
- Each row represents a sample from the same stratigraphic sequence, and
- There is a column named Depth containing depth/height for each sample.

Run calibration using the loaded age model. Three new columns will be appended:

- Age (mid)

- Upper 95% CI
- Lower 95% CI





Citation

Lu, Z.B., Pearson, P.N., Fan, J.X., et al. STRATIGIRAFFIC: A software tool for age–depth model development and application, with an application to ODP Site 1052 (in prep.).

Contact

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